MIL-C-13246B
17 October 1973
SUPERSEDING
MIL-C-13246A
7 June 1962

MILITARY SPECIFICATION

CRESYLIC ACID, TECHNICAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one technical grade of cresylic acid.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390 - Plastic, Molding and Extrusion Material, Polyethlene and Copolymers (Low, Medium, and High Density).
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-C-569 - Containers, Plastic, Molded (For Liquids, Pastes and Powders): Overpacked.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-G-432 - Gaskets-Nonmetallic, Synthetic Rubber.
MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.
MIL-C-52078 - Cap-Plug, Cap, and Plug, Protective, Plastic Dust and Moisture Seal.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171-179.")

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D95-70 - Water in Petroleum Products and Bituminous Materials by Distillation.
D255-70 - Steam Distillation of Bituminous Protective Coatings.
D850-70 - Distillation of Industrial Aromatic Hydrocarbons and Related Materials.
D1193-70 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Water content. Cresylic acid shall contain no more than 1.0 percent by weight of water when tested as specified in 4.2.4.1.

3.2 Distillation characteristics. Cresylic acid shall conform to the distillation characteristics of table I when tested as specified in 4.2.4.2.

Table I. Distillation characteristics

Percent by volume distilled	At temperature of (°C):	
	Minimum	Maximum
5.0	180	208
50.0	---	212
95.0	---	225
Dry point	---	230

3.3 Oil and naphthalene content. Cresylic acid shall contain no more than 0.25 percent by weight of oil and naphthalene when tested as specified in 4.2.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own

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or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Goverment. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the cresylic acid produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1 quart shall be removed from each sample packaging container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of packaging containers in batch or lot	Number of sample packaging containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
Over 7,000	13

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

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(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Palletization not as specified (when required)

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Water content. Determine the percent by weight of water in the specimen in accordance with ASTM D95.

4.2.4.2 Distillation characteristics. Determine the distillation characteristics of the specimen in accordance with ASTM D850.

4.2.4.3 Oil and naphthalene content. Assemble a steam distillation apparatus similar to that shown in ASTM D255 except insert a steam trap between the generator and the steam distillation flask to insure dry steam and use an adapter sufficiently long to reach the bottom of a 2-liter separatory funnel. Measure 100 milliliters (ml) of specimen into a 3-liter, ring-necked, round-bottom flask. Add 400 ml of sodium hydroxide solution having a specific gravity of 1.3 and agitate thoroughly. Place 50 ml of 5-percent sodium hydroxide solution in a 2-liter separatory funnel and place the funnel under the adapter in such a manner that the orifice of the adapter is below the surface of the sodium hydroxide solution. Connect the steam tube and start the distillation. Continue the distillation until the distillate is colorless, then remove the flask and without disturbing the position of the adapter, rinse the adapter and the condenser several times with ether. Remove the separatory funnel from the adapter and cool to below room temperature, then add 100 ml of ether and shake thoroughly. Release any pressure caused by the shaking by inverting the funnel and slowly opening the stopcock. Place the funnel in a cooling bath until a clear separation of the water and ether is apparent. Draw the water layer off into another 2-liter separatory funnel and again extract with 100 ml of ether. Combine both ether extracts and wash, first with 20 ml of 10-percent sodium hydroxide solution and then with 50-ml portions of water until the wash water is neutral to phenolphthalein indicator. Filter the washed ether and extract into a

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dry, tared 250-ml Erlenmeyer flask. Rinse the separatory funnel and filter pads several times with fresh ether and catch the rinsings in the flask. Place the flask on a steam bath and evaporate the ether (the ether may be recovered by use of a condenser). Place the flask and residue in an oven having a temperature of 90° ± 2° C for 45 minutes. Cool to room temperature in a desiccator and weigh. Calculate the percent by weight oil and naphthalene as follows:

$$\text{Percent oil and naphthalene} = \frac{A}{B}$$

where: A = Weight of residue in grams and
B = Specific gravity of specimen at 25°/15.5° C.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Cresylic acid shall be packaged level A or C as specified (see 6.2).

5.1.1 Level A, 1-gal container. A quantity of 8.6 pounds (lb) (+1.4 or -0 ounce) of cresylic acid shall be packaged in a nominal 1-gal capacity low density polyethylene container conforming to class B of PPP-C-569 within a type III box as specified therein. The closure for the polyethylene container shall be the nonintegral spout specified in PPP-C-569 except that the following restrictions shall apply. The spout shall be injection molded from polyethylene conforming to type I, class M, grade 2, category 3, color natural of L-P-390. The spout shall be designed for an extra length nominal 38-millimeter diameter screw cap having a minimum of three turns of buttress thread. The inner surface and top rim of the spout shall be sufficiently smooth and concentric to form a leakproof mating surface with a plug which shall be inserted after filling. A snug-fitting plug conforming to type I of MIL-C-52078 and formed from polyethylene conforming to type I, class optional, grade 2, category 3, color natural of L-P-390 shall be inserted in the opening of the spout after filling until the underside of the flange of the plug is flush with the rim of the spout. After the plug is inserted, there shall be no evidence of leakage in the exterior of the plastic container when positioned first upright and then inverted, before applying the screw cap. The plug shall be secured tightly in the spout by an extra length nominal 38-millimeter diameter, phenolic resin, buttress screw thread, linerless cap designed to fit the spout. The plastic container shall meet the special low temperature drop resistance requirement specified in PPP-C-569. The box shall be closed as specified in the appendix to PPP-C-569.

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5.1.2 Level C. A quantity of 8.6 lb (+1.4 or -0 ounce) or 440 (+3 or -0) lb as specified (see 6.2) of cresylic acid shall be packaged in accordance with any applicable Department of Transportation (DOT) regulations in a manner which will assure that the contents will be unaltered during shipment from the supply source to the first receiving activity and for a limited storage time.

5.2 Packing. Cresylic acid shall be packed level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-gal containers. Four 1-gal containers of cresylic acid, packaged as specified in 5.1.1, shall be packed upright in a snug-fitting box conforming to class 3, style 2, for a type 2 load not exceeding 85 lb of PPP-B-585. Fiberboard pads conforming to minimum grade W5c of PPP-F-320 shall be used as needed to prevent movement of the 1-gal containers within the box. The box shall be closed and strapped as specified in the appendix to PPP-B-585. Alternatively, at the option of the supplier, the four 1-gal containers shall be first packed in a fiberboard box meeting the minimum Uniform Freight Classification requirements, and then overpacked in the above specified wirebound wood box.

5.2.1.2 Fifty-five-gal container. A quantity of 440 (+3 or -0) lb of cresylic acid shall be packed in a nominal 55-gal capacity steel drum conforming to type I of PPP-D-729. All interior surfaces of the drum, including the steel closures, shall have a minimum of two coats of baked-on epoxy phenolic resin. The total thickness of the coating shall be 0.0015 $\pm$ 0.0002 inch. The drum shall be closed by tightening the vent plug to a torque of 15 to 18 pound-feet and by tightening the 2-inch diameter plug to a torque of 30 to 35 pound-feet. Gaskets for plugs shall be formed from ethylene propylene terpolymer and shall meet the requirements of type I or VI of MIL-G-432.

5.2.2 Level C. Uniform quantities of cresylic acid, packaged as specified in 5.1.2, shall be packed upright in a shipping container constructed to assure safe delivery to destination without leakage. The shipping container shall comply with any applicable DOT regulations and with Uniform Freight Classification rules or National Motor Freight Classification rules, as applicable. The container shall be sufficiently rigid to withstand transportation and to permit them to be stored in tiers at least 10 feet high without damage to containers or contents. The gross weight and size limitations of the container shall not exceed the tariff or classification limitation specified therein.

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5.3 Marking. Unit containers, shipping containers, and palletized loads shall be marked in accordance with any applicable DOT regulations and MIL-STD-129. In addition, each unit container shall be durably and legibly labeled to show the following precautionary information:

DANGER!
RAPIDLY ABSORBED THROUGH SKIN
CAUSES SEVERE BURNS

Do not get in eyes, on skin, on clothing.
Avoid breathing vapor.
Do not take internally.
In case of contact, immediately remove all contaminated clothing, including shoes, and flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention. Wash clothing before re-use.

5.4 Palletization. When specified in the contract or order (see 6.2), shipping boxes shall be palletized upright in accordance with the applicable requirements of MIL-STD-147. Pallets shall conform to the 40 by 48 inch style I, type I, class 1 of MIL-P-15011 and shall be treated with wood preservative as specified therein. Boxes shall be palletized upright in four tiers of six boxes per tier in accordance with the requirement for load type I of MIL-STD-147 using a wood cap over the load and lumber battens on the pallet deck to compress the load as specified in MIL-STD-147.

6. NOTES

6.1 Intended use. Cresylic acid is intended for use as a peptizing agent for aluminum soap thickened hydrocarbon fuel type flame and incendiary agents. This material is the recommended peptizer for use with M1 and M2 thickeners; for M4 thickener, it is recommended that 2-ethylhexanoic acid be used as the peptizer of choice.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit quantity required.
(c) Level of packaging required (see 5.1).
(d) Level of packing required (see 5.2).
(e) If palletization is required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Navy - AS
Air Force - 68

Review activity:

DSA-GS

User activities:

Army - SM
Navy - OS

Preparing activity:

Army - MU(EA)

Project No. 6810-0676

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
Attn: SMUEA-TS-S
Edgewood Arsenal, MD 21010

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